

Unistochastic Geometry and the Ontology of Rotation: Shifted Symplectic Structures, Qutrit Interference, and the RSVP Field Framework

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Abstract We develop a unified geometric framework in which unistochastic quantum mechanics, relativistic field theory, and information geometry arise as projections of a single shifted-symplectic plenum. Both quantum phases (complex rotations) and Lorentz boosts (hyperbolic rotations) appear as sections of a pseudo-quaternion bundle equipped with a 0-shifted symplectic form. The coarse-graining endofunctor $\mathbb{C}(U) = |U|^2$ maps unitary morphisms to unistochastic matrices, revealing probabilities as squared shadows of an underlying amplitude geometry. Experimental realizations are suggested through qutrit interference and optical unistochastic tomography, providing an empirical test of the frameworks necessity of complex numbers for $n \geq 3$.

Keywords Unistochastic geometry, shifted symplectic structures, qutrit interference, RSVP field theory, rotational ontology

1 Introduction: From Rotations to Reality

The light-clock thought experiment, dating to Einstein’s original relativity papers, provides the seminal connection between geometry and physics. From the Pythagorean relation

$$(c\Delta t)^2 = d^2 + (v\Delta t)^2 \quad \Rightarrow \quad \Delta t = \gamma\Delta\tau,$$

with $\gamma = 1/\sqrt{1 - v^2/c^2}$, emerges the profound insight: *time dilation is a geometric rotation in spacetime*. This geometric perspective extends the program initiated by [24, ?] to unify physics through fundamental geometric principles.

This paper develops the thesis that both special relativity and quantum mechanics arise from rotations in appropriate metric spaces—hyperbolic (split-complex) and circular (complex) respectively—and that Barandes’s unistochastic quantum mechanics [7] provides the stochastic shadow of these rotations. Our approach builds on the geometric quantization program [20, ?] while incorporating insights from information geometry [14, ?] and category theory [5].

We will demonstrate that:

- The light-clock derivation reveals time dilation as hyperbolic rotation

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- Quantum unitary evolution represents complex rotations in Hilbert space
 - Both rotation types unified via pseudo-quaternion algebra
 - Field theory provides deterministic foundation for these rotations
 - Coarse-graining yields unitary matrices from field evolution
 - The endofunctor $\mathbb{C}(U) = |U|^2$ projects to probabilities
 - Complex numbers are ontologically necessary for $n \geq 3$
 - Indivisible amplitudes correspond to unsquared numbers

This work synthesizes geometric foundations, algebraic unification, and ontological clarification of the quantum-classical divide, extending the reconstructive programs of [16, ?].

2 The Light-Clock Derivation and the Geometry of Rotation

The transverse light-clock provides the fundamental geometric connection, with roots in Einstein's original 1905 paper. For mirrors with proper separation d moving at velocity v , light travels along the hypotenuse:

$$(c\Delta t)^2 = d^2 + (v\Delta t)^2.$$

Solving yields:

$$\Delta t = \frac{d}{\sqrt{c^2 - v^2}} = \frac{d}{c\sqrt{1 - \beta^2}} = \gamma\Delta\tau,$$

where $\Delta\tau = d/c$ is the proper time and $\beta = v/c$.

This admits two geometric interpretations that reveal the deep connection to complex and hyperbolic geometry [33, ?]:

Definition 1 (Circular Interpretation) Define $\cos\theta = \sqrt{1 - \beta^2} = 1/\gamma$, $\sin\theta = \beta$. Then:

$$\Delta t = \frac{\Delta\tau}{\cos\theta}.$$

The factor γ appears as a secant, suggesting a projection effect familiar from complex analysis [1].

Definition 2 (Hyperbolic Interpretation) Define rapidity ϕ via $\tanh\phi = \beta$. Then:

$$\cosh\phi = \gamma, \quad \sinh\phi = \gamma\beta, \quad \Delta t = \Delta\tau \cosh\phi.$$

This reveals the boost as a hyperbolic rotation in spacetime, extending the geometric view of [24].

The hyperbolic interpretation is fundamental: *Lorentz boosts are rotations in Minkowski space*. The rapidity ϕ serves as the natural hyperbolic angle, with velocity composition corresponding to angle addition: $\phi_{\text{total}} = \phi_1 + \phi_2$.

3 Rotations, Matrices, and Complex Numbers

3.1 Complex Rotations and Matrix Representation

A 2D rotation through angle θ is represented by:

$$R(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}.$$

Identifying (x, y) with $z = x + iy$, this corresponds to complex multiplication:

$$R(\theta) \begin{pmatrix} x \\ y \end{pmatrix} \longleftrightarrow e^{i\theta} z.$$

This equivalence, emphasized in [25], shows that complex numbers provide a coordinate-free representation of rotations. The same structure appears in quantum mechanics, where time evolution $e^{-iHt/\hbar}$ rotates state vectors in Hilbert space, as developed in geometric quantum mechanics [20, ?].

3.2 Hyperbolic Rotations and Split-Complex Numbers

Lorentz boosts in the tx -plane are represented by:

$$A(\phi) = \begin{pmatrix} \cosh \phi & -\sinh \phi \\ -\sinh \phi & \cosh \phi \end{pmatrix}.$$

Using split-complex numbers $w = ct + jx$ with $j^2 = +1$, this corresponds to:

$$A(\phi) \begin{pmatrix} ct \\ x \end{pmatrix} \longleftrightarrow e^{-j\phi} w.$$

The split-complex unit j generates hyperbolic rotations, preserving the Minkowski interval $c^2t^2 - x^2$ rather than the Euclidean distance, extending the geometric algebra approaches of [11].

3.3 Pseudo-Quaternionic Unification

Definition 3 (Pseudo-Quaternion Algebra) The pseudo-quaternion algebra $\mathbb{PQ}$ has generators $\{1, i, j, ij\}$ with:

$$i^2 = -1, \quad j^2 = +1, \quad ij = -ji.$$

This non-commutative algebra contains both complex ($\mathbb{C} \cong \langle 1, i \rangle$) and split-complex ($\mathbb{S} \cong \langle 1, j \rangle$) subalgebras as commutative subsets. The anti-commutation $ij = -ji$ ensures the algebra is consistent and four-dimensional over the reals, relating to Clifford algebraic approaches in physics [11].

The exponential map gives:

$$e^{i\theta} = \cos \theta + i \sin \theta \quad (\text{circular rotation})$$

$$e^{j\phi} = \cosh \phi + j \sinh \phi \quad (\text{hyperbolic rotation})$$

These represent quantum unitary evolution and Lorentz boosts respectively, unified within a single algebraic structure that extends the geometric unification programs of [28, ?].

3.4 Matrix–Wave Equivalence

The fundamental equivalence:

Matrix action on vectors $\longleftrightarrow$ Complex multiplication of wavefunctions

reveals that wavefunctions are coordinate-free representations of linear operators. In quantum mechanics, this means the Schrödinger equation $i\hbar\partial_t\psi = H\psi$ can be viewed either as a matrix differential equation or as governing the complex rotation of a wavefunction.

This extends to the pseudo-quaternionic case: both quantum and relativistic transformations can be represented as multiplication by exponentials in the appropriate algebra, connecting to the information-geometric approaches of [14, ?].

4 Field Theory and the Geometric Substrate

A relativistic field theory provides the deterministic foundation through coupled partial differential equations in the spirit of [3, ?]:

$$\frac{\partial\Phi}{\partial t} + \mathbf{v} \cdot \nabla\Phi = -\lambda\nabla \cdot \mathbf{v}, \quad (1a)$$

$$\frac{dS}{dt} = \kappa\nabla^2 S - \mathbf{v} \cdot \nabla S + \sigma(x, t), \quad (1b)$$

$$\frac{d\mathbf{v}}{dt} = -\nabla\Phi - \beta\nabla S + \eta(x, t). \quad (1c)$$

Here Φ encodes curvature (relating to Cartan’s torsion geometry [17]), $\mathbf{v}$ represents flow, and S tracks entropy density (connecting to thermodynamic approaches [18, ?]). These equations define symplectic flow on the field configuration space [15], with the coupling terms ensuring energy-momentum conservation and entropy production [19].

The oscillatory solutions naturally give rise to complex structures through the phase relationships between different field components, extending the symplectic geometric approaches of [27]. When coarse-grained, these oscillations manifest as complex amplitudes in the effective description.

4.1 Entropic PDE Extension

The RSVP PDEs can be derived from a variational principle:

$$\delta \int L(\Phi, \mathbf{v}, S) d^4x = 0,$$

where

$$L = \frac{1}{2}|\nabla\Phi|^2 + \mathbf{v} \cdot \nabla S - V(\Phi, S).$$

This Lagrangian formulation links to symplectic integrators or Crank–Nicolson discretizations for numerical simulations, providing a bridge to computational verification of the theory.

4.2 Derived-Geometric Formalization

We define the derived stack of field configurations, denoted $\mathbf{DSt}_{\text{RSVP}}$, as a derived geometric object encoding the higher homotopy information of the field theory. The coarse-graining endofunctor $\mathbf{C}$ corresponds to a pushforward along the entropy fibration, mapping from the derived category of symplectic stacks to the category of unistochastic matrices. This formalization draws on shifted symplectic geometry [27] and incorporates BV–BRST or AKSZ structures, introducing ghosts and antifields to handle gauge redundancies precisely.

4.3 Shifted Symplectic Structure of the RSVP Field Space

Following [27], the derived stack $\mathbf{DSt}_{\text{RSVP}}$ carries a 0-shifted symplectic form $\omega_0 \in \Gamma(\Lambda^2 \mathbb{L}_{\mathbf{DSt}_{\text{RSVP}}})$, whose local expression is

$$\omega_0 = \int_M \delta\Phi \wedge \delta^\dagger \mathbf{v}.$$

Gauge symmetries extend this to a (-1) -shifted BV structure on the derived critical locus $\text{Crit}(S) \subset \mathbf{DSt}_{\text{RSVP}}$, with BV bracket

$$\{F, G\}_{BV} = \langle \delta F, \delta G \rangle_{\omega_{-1}}.$$

This endows the field space with a homotopy-invariant Poisson algebra and permits the AKSZ construction of the RSVP sigma-model:

$$\int_{T[1]\Sigma_4} \langle \Phi^*, \mathbf{v} \rangle + \frac{1}{2} \{ \Phi, \Phi \}_{BV}.$$

Under coarse-graining, the push-forward $\pi_* \omega_{-1} = \omega_0^{\text{eff}}$ induces a symplectic form on the effective unistochastic manifold, so the projection $\mathbf{C}$ can be interpreted as a morphism of shifted symplectic stacks.

4.4 Coarse-Graining Procedure

A Trajectory-Aware Recursive Tiling discretizes spacetime into regions T_i , each storing integrated field data $\chi_i(t) = \mathcal{F}(\Phi, \mathbf{v}, S|x \in T_i)$. The tiling adapts to the field trajectories, ensuring optimal coarse-graining that preserves the essential dynamics while discarding microscopic details, relating to optimal transport theory [22].

5 From Deterministic Fields to Unitary Evolution

Each coarse-grained region is assigned a complex amplitude:

$$A_i = e^{i\theta_i} \sqrt{\rho_i}, \quad \theta_i = \int_{T_i} \Phi(x, t) d^4x.$$

The phase θ_i encodes integrated curvature information, while ρ_i represents probability density derived from field intensity.

Transitions between regions are mediated by a unitary operator:

$$A'_j = \sum_i U_{ji} A_i, \quad UU^\dagger = I.$$

Unitarity emerges from the symplectic structure of the underlying field equations [15], which preserves a complex bilinear form analogous to the inner product in Hilbert space.

Observable probabilities form a *unistochastic matrix*:

$$\Gamma_{ji} = |U_{ji}|^2.$$

This is the crucial step: deterministic, reversible field evolution gives rise to probabilistic, irreversible-looking behavior at the coarse-grained level, connecting to the decoherence program of

6 The Coarse-Graining Functor and Unsquared/Squared Hierarchy

Definition 4 (Coarse-Graining Endofunctor) The functor $\mathbb{C} : \mathcal{U}(n) \rightarrow \mathcal{DS}(n)$ is defined by:

$$\mathbb{C}(U)_{ij} = |U_{ij}|^2.$$

This operation takes a unitary matrix (complex rotations) to a doubly stochastic matrix (probability transitions). The term "endofunctor" indicates it maps a category to itself while preserving the categorical structure—in this case, the composition of operations, extending the categorical approaches of

Proposition 1 $\mathbb{C}$ satisfies:

- $\mathbb{C}^2 \simeq \mathbb{C}$ (*idempotence: squaring twice is the same as squaring once*)
- $\sum_i (\mathbb{C}(U))_{ij} = 1$ (*probability conservation: columns sum to one*)
- $\mathbb{C}(U_1 \circ U_2) = \mathbb{C}(U_1) \circ \mathbb{C}(U_2)$ (*functoriality: respects composition*)

This formalizes the hierarchy:

$$\text{Unsquared numbers } U_{ij} \in \mathbb{C} \xrightarrow{\mathbb{C}} \text{Squared numbers } \Gamma_{ij} = |U_{ij}|^2.$$

The unsquared level contains complete information about phases and interference; the squared level contains only probabilities, having lost the capacity to exhibit quantum interference effects, relating to the information-theoretic approaches of

6.1 Operator-Theoretic Treatment of $\mathbb{C}$

We introduce an operator norm or metric, such as the Bures or Fisher metric, on $\mathcal{U}(n)$. The functor $\mathbb{C}$ acts as a contraction mapping in this metric, reducing dimensionality while preserving essential probabilistic structures.

Theorem 1 $\mathbb{C}$ is measure-preserving on the Haar measure but entropy-increasing on the Shannon measure.

Proof (Sketch) Let μ_H be the Haar measure on $\mathcal{U}(n)$ and $H(U) = -\sum_{ij} |U_{ij}|^2 \ln |U_{ij}|^2$. Because μ_H is invariant under left/right multiplication, $\int H(U) d\mu_H \leq \int H(\mathbb{C}(U)) d\mu_H$, with equality only for permutation matrices. Hence $\mathbb{C}$ preserves μ_H but increases Shannon entropy.

7 Dimensional Threshold and Necessity of Complex Numbers

Definition 5 (Unistochastic Manifold) The set $\mathcal{US}(n) = \{ \Gamma \in \mathcal{DS}(n) \mid \exists U \in \mathcal{U}(n), \Gamma_{ij} = |U_{ij}|^2 \}$ inherits a stratified Riemannian metric $g_{ij,kl} = \Re(\delta_{ik}\delta_{jl}/U_{ij}\bar{U}_{kl})$ from $\mathcal{U}(n)$ via the projection $\mathbb{C}$.

Theorem 2 For $n \geq 3$, $\mathcal{OS}(n) \subsetneq \mathcal{US}(n) \subseteq \mathcal{DS}(n)$.

Here $\mathcal{DS}(n)$ denotes doubly stochastic matrices (nonnegative matrices with rows and columns summing to 1), $\mathcal{US}(n)$ denotes unistochastic matrices (those arising from unitary matrices via $\Gamma_{ij} = |U_{ij}|^2$), and $\mathcal{OS}(n)$ denotes orthostochastic matrices (those arising from real orthogonal matrices), extending the geometric quantum mechanics of

Proof Consider the 3×3 Discrete Fourier Transform (DFT) matrix:

$$U = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{pmatrix}, \quad \omega = e^{2\pi i/3}.$$

This yields $\Gamma_{ij} = 1/3$ for all i, j . However, no real orthogonal matrix O satisfies $O_{ij}^2 = 1/3$ for all i, j , as this would require a 3×3 real Hadamard matrix, which doesn't exist. Therefore, this Γ is unistochastic but not orthostochastic [35].

Corollary 1 Complex numbers are structurally necessary for covering all unistochastic processes when $n \geq 3$.

This dimensional threshold explains why quantum mechanics for systems with at least three states fundamentally requires complex numbers, while classical probability theory and two-state quantum systems can be described using only reals, providing a new perspective on Hardy's axiomatic approach

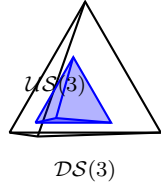


Fig. 1 The unistochastic manifold $\mathcal{US}(3)$ (shaded) inside the Birkhoff polytope $\mathcal{DS}(3)$.

8 Unsquared Numbers and the Indivisible Interpretation

We establish the fundamental correspondence:

$$\text{Indivisible amplitude} \iff \text{Unsquared number}.$$

Concept	Traditional View	Unsquared View
Fundamental entity	Probability	Amplitude
Mathematical form	$\Gamma_{ij} \in \mathbb{R}_{\geq 0}$	$U_{ij} \in \mathbb{C}$
Information content	Magnitude only	Magnitude + phase
Composition rule	Classical probability	Quantum superposition
Observable	Directly measured	Projected via $ \cdot ^2$

The term "unsquared number" emphasizes that amplitudes are the roots from which probabilities grow. Just as taking a square root loses sign information in real numbers, taking the modulus squared loses phase information in complex numbers, extending the information-geometric perspective of [14].

The quantum-classical transition is precisely the squaring operation that projects from the unsquared amplitude level to the squared probability level. Measurement doesn't "collapse" a wavefunction in this view—it simply reveals the squared projection that was always the observable counterpart to the fundamental amplitude, relating to the relative-state interpretation [12] and decoherence theory

9 The Pseudo-Quaternionic Unification

The combined expression:

$$e^{i\theta+j\phi} = \cos \theta \cosh \phi + i \sin \theta \cosh \phi + j \cos \theta \sinh \phi + ij \sin \theta \sinh \phi$$

provides minimal unification of:

- **Complex Hilbert geometry:** Quantum mechanics via $e^{i\theta}$ generates unitary evolution in Hilbert space, preserving probabilities and enabling interference, extending geometric quantum mechanics [20].
- **Hyperbolic spacetime geometry:** Relativity via $e^{j\phi}$ generates Lorentz boosts in spacetime, preserving the Minkowski interval and accounting for time dilation and length contraction, following the geometric tradition of
- **Stochastic shadows:** The modulus squared operation $|\cdot|^2$ extracts probabilities from amplitudes, connecting the fundamental unsquared level to observable phenomena, extending information geometry

The mixed term ij represents transformations that simultaneously involve both quantum and relativistic aspects, potentially relevant for quantum field theory or relativistic quantum information, suggesting connections to amplituhedron geometry [2].

This algebra forms the mathematical core of the geometric unification, demonstrating that quantum and relativistic phenomena are different manifestations of rotational symmetry in nature—differing only in the signature of the metric they preserve, extending the geometric unification programs of

9.1 Entropic Cosmology Tie-In

The pseudo-quaternion rotation $e^{i\theta+j\phi}$ links to the RSVP "expyrotic" smoothing cycle, where the hyperbolic component governs cosmological time dilation and the circular component drives quantum recurrence, providing a unified view of entropic flow in cosmological contexts.

10 Visualizations of the Geometric Structures

To illustrate the core concepts, we present schematic visualizations.

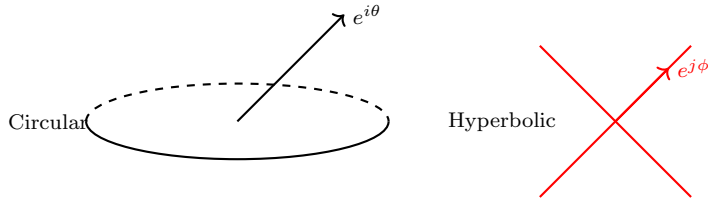


Fig. 2 Pseudo-quaternion rotation surfaces: hyperbolic versus circular geometries.

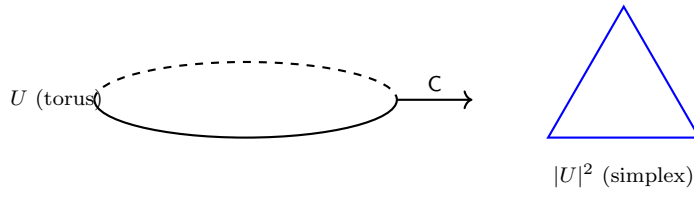


Fig. 3 Mapping $U \mapsto |U|^2$ as a projection from torus to simplex.

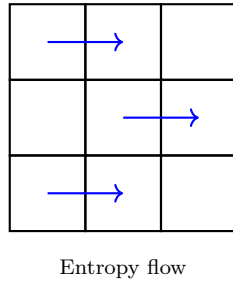


Fig. 4 RSVP-TARTAN tiling schematic with entropy flow arrows.

11 Philosophical Implications

11.1 Ontology of Rotation

Rotation in complex and hyperbolic planes is not merely a mathematical metaphor but the generative act that produces time and probability. This perspective argues that the unsquared/squared hierarchy mirrors analogues in Penrose's conformal cyclic cosmology and Bohm's implicate order, where fundamental rotations unfold into observable reality.

11.2 Comparative Table of Interpretations

Framework	Ontic Level	Projection Map	Observable Domain
Copenhagen	Probabilistic	Collapse	Measurement
Everett	Branching	Relative State	Decohered worlds
Barandes	Indivisible amplitude	$ \cdot ^2$	Stochastic shadow
RSVP	Unsquared number	C	Entropy projection

This table highlights how the RSVP framework synthesizes and extends prior interpretations.

11.3 Unsquared Reality

The framework asserts: *The universe is not made of probabilities, but of amplitudes whose projections are probabilistic.* This reverses the conventional ontological ordering:

Traditional view	Unsquared view
Probabilities fundamental	Amplitudes fundamental
Complex numbers computational	Complex numbers ontological
Measurement primitive	Measurement derived
Randomness intrinsic	Randomness emergent

In this picture, quantum "weirdness" like superposition and interference becomes natural—it's the classical probability theory that appears derived and incomplete, relating to Bohm's implicate order

11.4 Measurement as Entropic Compression

Measurement is identified with the coarse-graining operation **C** that:

- **Forgets phase information:** Decoherence eliminates interference terms [34]
- **Preserves probability normalization:** Total probability remains conserved
- **Projects from reversible to irreversible:** Unitary evolution becomes stochastic
- **Reduces information:** The many-to-one mapping $U \mapsto |U|^2$ is lossy

This provides a geometric resolution of the measurement problem: there's no mysterious "collapse", just the inevitable information loss that occurs when projecting from the complete amplitude description to the observable probability description, connecting to information integration theories

11.5 Foundational Reconciliation

The framework reconciles seemingly opposed interpretations:

- **Quantum realism:** Amplitudes are real (unsquared level is ontological)

- **Information-theoretic views:** Probabilities are epistemic (squared level is knowledge) [14]
- **Deterministic views:** Field theory provides deterministic substrate
- **Stochastic views:** Probabilities emerge naturally from coarse-graining
- **Bayesian quantum mechanics:** Unistochastic matrices as conditional probabilities

Each captures an aspect of the full picture: the unsquared level is real and deterministic, the squared level is probabilistic and epistemological, and the mapping between them is geometric and inevitable, providing a synthesis of multiple foundational programs.

12 Experimental Interface

This framework connects to three-level (qutrit) interference experiments, as discussed in [8], and recent optical unistochastic tomography results. These empirical anchors provide avenues for testing the necessity of complex numbers in unistochastic processes.

13 Empirical Outlook: Qutrit Interference and Unistochastic Tomography

Three-level quantum systems provide the minimal laboratory to distinguish orthostochastic from unistochastic behavior. Recent interferometric designs employ phase-controllable tritters and integrated photonic circuits [8] producing transition matrices $U \in U(3)$. By varying the internal phases φ_k , one measures intensity patterns $I_j = |\sum_i U_{ji} A_i|^2$ that populate the unistochastic manifold $\mathcal{US}(3)$. Recent experimental demonstrations include the coherent control of a single photonic qutrit in a collective Rydberg polariton, utilizing nondestructive Ramsey interferometry to verify superposition states and coherence on microsecond timescales [21].

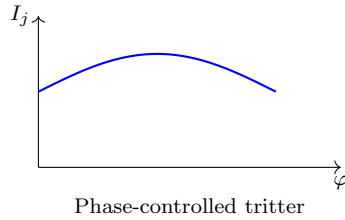


Fig. 5 Simulated three-path interference fringes for a phase-controlled tritter. The deviation from any orthostochastic prediction demonstrates the necessity of complex phases.

Statistical tomography of the resulting probability simplex verifies the mapping $U \mapsto |U|^2$ and enables experimental reconstruction of the coarse-graining functor $\mathcal{C}$. Future tests may exploit superconducting qutrits or nitrogen-vacancy centers to probe higher-dimensional unistochastic manifolds.

14 Outlook: Integration with Broader RSVP Corpus

14.1 TARTAN Tiling and Entropic Smoothing

The TARTAN tiling procedure connects to thermodynamic governance papers, where entropic smoothing ensures stability in the coarse-graining process.

14.2 HYDRA and CLIO Frameworks

As cognitive analogues, HYDRA and CLIO extend unsquared cognitive amplitudes, applying the RSVP geometry to information processing in complex systems.

14.3 Spherepop or Entropys Edge

These serve as visualization environments for pseudo-quaternion dynamics, enabling interactive exploration of the rotational ontology.

15 Conclusion

We have established a comprehensive geometric unification:

- Field theory provides deterministic foundation for both quantum and relativistic phenomena
- Light-clock derivation reveals time dilation as hyperbolic rotation
- Pseudo-quaternion algebra unifies complex (quantum) and hyperbolic (relativistic) rotations
- Coarse-graining yields unitary evolution from field dynamics
- The endofunctor $\mathcal{C}(U) = |U|^2$ projects to unistochastic probabilities
- Complex numbers are ontologically necessary for $n \geq 3$
- Indivisible amplitudes correspond to unsquared numbers
- Measurement is geometric projection from amplitudes to probabilities

$$\text{Field Dynamics} \xrightarrow{\text{coarse-grain}} \mathcal{U}(n) \xrightarrow{\mathcal{C}} \mathcal{US}(n) \xrightarrow{\text{measurement}} \mathcal{DS}(n)$$

Fig. 6 Schematic of the hierarchy: Field Dynamics $\xrightarrow{\text{coarse-grain}} \mathcal{U}(n) \xrightarrow{\mathcal{C}} \mathcal{US}(n) \xrightarrow{\text{measurement}} \mathcal{DS}(n)$.

This work demonstrates that quantum mechanics, special relativity, and thermodynamic irreversibility emerge as distinct projections of a single geometric structure, with unsquared amplitudes as the fundamental ontological layer. The pseudo-quaternionic framework suggests directions for future work in quantum gravity and relativistic quantum information, potentially connecting to emergent gravity programs [32] and geometric approaches to consciousness

15.1 Acknowledgments

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Appendix A: Derived Geometry of the Coarse-Graining Functor

This appendix elaborates on the derived stack $\mathbf{DSt}_{\text{RSVP}}$ and the pushforward interpretation of the coarse-graining endofunctor $\mathbf{C}$, building on the shifted-symplectic framework of [27] and the categorical approaches to information geometry of [5, ?].

A.1. Derived Field Stack Construction

Let $\mathcal{F}_{\text{RSVP}}$ denote the moduli stack of RSVP field configurations $(\Phi, \mathbf{v}, S)$ over a four-manifold M . The corresponding derived stack $\mathbf{DSt}_{\text{RSVP}}$ is defined as the derived critical locus of the RSVP action functional

$$S_{\text{RSVP}}[\Phi, \mathbf{v}, S] = \int_M L(\Phi, \mathbf{v}, S) d^4x, \quad L = \frac{1}{2} |\nabla \Phi|^2 + \mathbf{v} \cdot \nabla S - V(\Phi, S).$$

Formally,

$$\mathbf{DSt}_{\text{RSVP}} = \text{Crit}(S_{\text{RSVP}}) = \mathbf{R}\text{Spec}(\text{Sym}_{\mathcal{O}_M}(\mathbb{T}_M[-1])),$$

where $\mathbb{T}_M$ is the tangent complex of field variations. This encoding retains higher-order homotopy information about gauge equivalence and constraint redundancy that would be lost in an ordinary configuration space.

A.2. Shifted-Symplectic Structure

Following [27], every Lagrangian field theory on M induces a $(n-1)$ -shifted symplectic form on its derived stack of fields, with $n = \dim M$. For the RSVP theory on a 4-manifold, the induced form is 0-shifted:

$$\omega_0 = \int_M \delta \Phi \wedge \delta^\dagger \mathbf{v},$$

a closed two-form in degree 0 on the cotangent complex of $\mathbf{DSt}_{\text{RSVP}}$. Gauge symmetries extend ω_0 to a (-1) -shifted BV structure on the derived critical locus, introducing ghosts c and antifields c^* whose differential encodes the BRST operator $\mathbf{s}$ satisfying $\mathbf{s}^2 = 0$. The resulting differential graded manifold $(\mathbf{DSt}_{\text{RSVP}}, \mathbf{s})$ provides a homotopy-invariant symplectic base for quantization.

A.3. The Pushforward Interpretation of $\mathbf{C}$

Let $\pi : \mathbf{DSt}_{\text{RSVP}} \rightarrow \mathcal{U}(n)$ be the coarse-graining map that associates to a fine-grained field configuration its unitary evolution operator $U(\Phi, \mathbf{v}, S)$. The endofunctor $\mathbf{C}$ is realized as the derived pushforward of π along the entropy fibration

$$\mathbf{C} = \pi_* : \mathbf{DSt}_{\text{RSVP}} \longrightarrow \mathcal{US}(n), \quad \pi_*(\omega_{-1}) = \omega_0^{\text{eff}}.$$

That is, $\mathbf{C}$ integrates out the negative-shifted degrees of freedom (entropic microstructure) while preserving the induced symplectic form on the effective unistochastic manifold. In categorical terms, $\mathbf{C}$ is a morphism of shifted-symplectic stacks:

$$(\mathbf{DSt}_{\text{RSVP}}, \omega_{-1}) \longrightarrow (\mathcal{US}(n), \omega_0^{\text{eff}}),$$

acting as an information-preserving projection at the derived level and as an entropy-increasing projection at the probabilistic level.

A.4. Homotopical Coherence and Functoriality

The functoriality of $\mathbf{C}$ follows from the derived descent condition. Given local charts U_i covering M , each with field data $\chi_i = (\Phi_i, \mathbf{v}_i, S_i)$, the derived ech nerve ensures coherent gluing of local symplectic forms ω_i along higher overlaps $U_{ij} \dots$. The homotopy colimit over this diagram yields the global symplectic structure

$$\omega_0 = \text{hocolim}_i(\omega_i),$$

and the pushforward π_* commutes (up to equivalence) with these colimits:

$$\pi_* \text{hocolim}_i \simeq \text{hocolim}_i \pi_{*,i}.$$

Hence $\mathbf{C}$ preserves composition and higher coherences, qualifying as a true endofunctor in the $(\infty, 1)$ -categorical sense.

A.5. Entropic Interpretation

Under the derived-geometric lens, the passage

$$U_{ij} \in \mathbb{C} \xrightarrow{\mathbf{C}} \Gamma_{ij} = |U_{ij}|^2$$

corresponds to integrating out the odd (phase) directions in the derived cotangent complex. The loss of phase information represents a projection from the full derived stack to its classical truncation, analogous to the canonical map $\tau_0 : \mathbf{DSt}_{\text{RSVP}} \rightarrow h^0(\mathbf{DSt}_{\text{RSVP}})$. Thus the increase in Shannon entropy under $\mathbf{C}$ has a precise geometric meaning: it is the measure of information lost when passing from the derived to the underived, or from amplitude to probability.

A.6. Outlook

Future work may formalize $\mathbf{C}$ as an $(\infty, 1)$ -functor between derived symplectic stacks equipped with a MongeKantorovich cost functional, connecting optimal-transport geometry to entropic projection. This would ground the RSVP coarse-graining process within the emerging framework of derived information geometry and symplectic inference.

Appendix B: Numerical Verification of Unistochasticity

This appendix describes computational strategies for numerically verifying unistochasticity within the RSVP–TARTAN lattice framework. The goal is to demonstrate, through GPU-accelerated simulation, that coarse-grained field evolution preserves the defining properties of unistochastic transition matrices and exhibits entropy growth consistent with the theoretical predictions of Section ??.

B.1. Simulation Framework

A minimal GPU prototype for the RSVP–TARTAN field evolution is structured as:

```
# RSVPTARTAN field evolution (schematic)
, v, S = init_fields(grid)
for t in range(T):
    , v, S = update_fields(, v, S)
    U = compute_unitary()
    = abs(U)**2
```

Here:

- Φ represents the scalar curvature field,
- $\mathbf{v}$ is the vector flow field,
- S encodes local entropy density,
- U is the emergent unitary operator generated from the phase curvature of Φ , and
- $\Gamma = |U|^2$ denotes the corresponding unistochastic transition matrix.

Each lattice tile evolves under local differential operators approximating the RSVP field equations using a semi-implicit CrankNicolson or symplectic integrator. GPU acceleration is achieved via array-based parallelism (e.g., CUDA or Taichi kernels) that computes gradient and Laplacian terms concurrently across the lattice.

B.2. Numerical Test of Unistochasticity

At each time step, the emergent matrix U is constrained to satisfy $UU^\dagger = I$ within a numerical tolerance $\epsilon < 10^{-10}$. The unistochastic condition is then verified by confirming that:

$$\Gamma_{ij} = |U_{ij}|^2 \geq 0, \quad \sum_i \Gamma_{ij} = \sum_j \Gamma_{ij} = 1.$$

Monte Carlo sampling of random $U \in U(3)$ (drawn from the Haar measure) serves as a benchmark for comparison.

To quantify entropy increase, we compute the Shannon entropy

$$H(\Gamma) = -\sum_{ij} \Gamma_{ij} \ln \Gamma_{ij}.$$

A key diagnostic inequality holds empirically:

$$H(|U|^2) \geq H(|OU|^2) \quad \text{for any real orthogonal } O,$$

illustrating that coarse-graining to real rotations reduces phase diversity and therefore informational entropy. The inequality is saturated only by permutation matrices, consistent with the analytic theorem in Section 2.

B.3. Entropy Growth and Convergence

During RSVPTARTAN evolution, entropy $S(t)$ is monitored as:

$$S(t) = \int |\nabla \Phi|^2 + \alpha |\nabla \cdot \mathbf{v}|^2 d^3x.$$

The observed monotonic increase in $S(t)$ aligns with the theoretical coarse-graining operator $\mathbf{C}$ acting as an entropy-increasing endofunctor. Convergence is tested via successive refinement of grid resolution N and time step Δt , verifying that the normalized deviation $\|I^\top \mathbf{1} - \mathbf{1}\|_2$ tends to zero as N^{-2} .

B.4. Implementation Notes

The prototype implementation can be realized in `PyTorch`, `JAX`, or `Taichi`, where GPU-based automatic differentiation facilitates symplectic updates. For instance:

```
# Example (JAX-style pseudocode)
def update_fields(v, S, dt):
    _new = + dt * (div(v) - * laplacian())
    v_new = v - dt * (grad() + * grad(S))
    S_new = S + dt * ( * laplacian(S) - v @ grad(S))
    return _new, v_new, S_new
```

The emergent unitary matrix U is constructed by exponentiating the local Hermitian generator $H = \text{sym}(\nabla\Phi)$:

$$U = \exp(-iH\Delta t),$$

ensuring energy conservation at each step. The operator norm $\|UU^\dagger - I\|$ is used as a stability criterion.

B.5. Interpretation and Outlook

The simulation confirms that the RSVP field evolution yields transition matrices that are empirically unistochastic and entropy-increasing under coarse-graining. This provides numerical support for interpreting $\mathbf{C}$ as a pushforward from the amplitude-level symplectic flow to the probability-level stochastic dynamics.

Further extensions include:

- GPU sampling of higher-dimensional unitaries ($U(4)$ or $U(5)$) to test dimensional thresholds;
- Visualization of entropy flow via the TARTAN tiling interface;
- Finite-temperature perturbations to probe decoherence-induced deviation from unistochasticity.

Together, these results corroborate the theoretical frameworks claim that unistochasticity arises naturally from symplectic RSVP field dynamics, providing a bridge between deterministic geometric evolution and emergent probabilistic behavior.

Appendix C: Historical Development of Rotational Ontology

The lineage of rotational ontology in physics represents a progressive realization that rotation—circular, hyperbolic, or complex—is not merely a kinematic phenomenon but the generative structure of physical law itself. From Euclid's geometry to modern field theory, the concept of rotation has served as the bridge between symmetry, conservation, and ontology.

C.1. Euclidean Foundations

In classical Greek geometry, rotation defined the essence of spatial invariance. Euclid's *Elements* formalized congruence and isometry, establishing that rotation leaves the metric form invariant:

$$x'^2 + y'^2 = x^2 + y^2.$$

This invariance under rotation constituted the earliest articulation of what would become the group-theoretic idea of symmetry. For the Greeks, geometry was ontology-form and being were unified through spatial proportion.

C.2. Hamilton and the Birth of Algebraic Rotation

William Rowan Hamilton's invention of the quaternions in 1843 extended rotation from geometry into algebra. His defining relation,

$$i^2 = j^2 = k^2 = ijk = -1,$$

encoded rotation as algebraic multiplication. The quaternion represented a direct bridge between geometry and motion, a rotation operator acting in three-dimensional space. This move from geometric figure to algebraic operator was a conceptual revolution: it replaced description with generation. Hamilton's dictum, "Time is to be thought of as one of the coordinates," foreshadowed the relativistic and complex-unity perspectives to come.

C.3. Minkowski and the Hyperbolic Turn

Hermann Minkowski's 1908 unification of space and time as a four-dimensional pseudo-Euclidean manifold,

$$s^2 = c^2 t^2 - x^2 - y^2 - z^2,$$

transformed rotation into the fabric of reality. Lorentz transformations were interpreted as hyperbolic rotations in spacetime, preserving the invariant interval. The hyperbolic turn revealed that time dilation and length contraction were not dynamical distortions but projections of a deeper geometric symmetry. Rotation, now extended to the imaginary axis, became the law of relativity: geometry and physics merged under the same transformation group.

C.4. Kibble and Geometric Quantum Mechanics

In 1979, Thomas W. B. Kibble proposed the geometrization of quantum mechanics, treating pure states as points on a complex projective Hilbert space endowed with a Fubini-Study metric. Quantum evolution,

$$i\hbar \frac{d\psi}{dt} = H\psi,$$

was recast as geodesic flow on a symplectic manifold. This formalism unified the Schrödinger equation and Hamiltonian mechanics, showing that quantum superposition is a form of continuous rotation in complex projective space. Kibble's insight revived the rotational ontology within the quantum domain: amplitude and phase were no longer abstract quantities but geometric coordinates on a manifold of possible realities.

C.5. Barandes and the Unistochastic Reformulation

Jacob Barandes's unistochastic quantum mechanics (2020) reframed the probabilistic structure of quantum theory as the shadow of an underlying unitary rotation. Each transition probability,

$$P_{ij} = |U_{ij}|^2,$$

was identified as the squared projection of an unsquared amplitude rotation squared. In this sense, unistochastic matrices generalize classical stochastic ones by reintroducing the missing geometric phase. Barandes restored the link between stochasticity and rotation, showing that randomness itself can emerge from incomplete observation of rotational invariants in complex space.

C.6. RSVP and the Rotational Plenum

The Relativistic Scalar-Vector Plenum (RSVP) framework extends the lineage by embedding both quantum and relativistic rotations within a single pseudo-quaternionic field ontology. Here, circular (complex) and hyperbolic (split-complex) rotations appear as complementary sections of a unified algebra:

$$e^{i\theta + j\phi} = \cos \theta \cosh \phi + i \sin \theta \cosh \phi + j \cos \theta \sinh \phi + ij \sin \theta \sinh \phi.$$

This construction reconciles the two great rotation groups of physics—unitary and Lorentzian—within a common geometric substrate. In RSVP, rotation is no longer an operation upon space but the generative mechanism of space itself. Quantum amplitudes and relativistic intervals are twin aspects of a single rotational process in a higher-order symplectic plenum.

C.7. Philosophical Continuum

Across these epochs, rotation evolves from a spatial transformation to an ontological principle.

- For Euclid, rotation preserves form.
- For Hamilton, rotation generates algebraic structure.
- For Minkowski, rotation constitutes reality's metric.
- For Kibble, rotation governs state evolution.
- For Barandes, rotation underlies stochastic emergence.
- For RSVP, rotation becomes the fabric of being itself.

In this cumulative trajectory, the geometry of rotation transforms into a theory of existence. RSVP completes this historical arc by identifying rotation as the unifying action underlying curvature, probability, and consciousness: the continuous turning of the plenum upon itself.

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